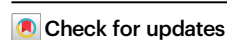


A universal foundation model for grounded biomedical image interpretation

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The integration of AI-assisted biomedical image analysis into clinical practice demands AI-generated findings that are not only accurate but also interpretable. However, existing models generally lack the ability to simultaneously generate diagnostic findings and localize corresponding targets. This limitation makes it challenging to correlate AI-generated findings with visual evidence and interpret the results. To this end, we introduce UniBiomed, a universal foundation model for grounded biomedical image interpretation, which is capable of generating accurate diagnostic findings and segmenting the biomedical targets. UniBiomed is based on an integration of Multi-modal Large Language Model and Segment Anything Model, which can unify diverse biomedical tasks in universal training for advancing grounded interpretation. To develop UniBiomed, we curate a large-scale dataset comprising 27 million triplets of images, region annotations, and text descriptions. Extensive validation on 70 internal and 14 external datasets demonstrated the state-of-the-art performance of UniBiomed in diverse biomedical tasks.

Multi-modal interpretation of biomedical images opens up novel opportunities in biomedical image analysis^{1–3}. Visual information from biomedical imaging enables detailed anatomical and functional analysis from cell to organ levels^{4–11}, while the textual information from diagnostic findings provides fine-grained descriptions for interpreting imaging^{12–14}. However, it remains challenging for existing biomedical AI models to effectively integrate holistic vision and language information to assist clinicians in clinical practice. Although recent multi-modal biomedical foundation models^{14–19} have showcased encouraging results in interpreting biomedical images and generating

diagnostic findings, these models are typically region-agnostic and fail to extract the target regions (e.g., tiny lesions) described in the generated findings. This limitation poses a critical barrier for clinicians to associate AI-derived findings with specific regions in images and interpret the results, significantly hindering the clinical practice of biomedical AI models.

Biomedical image segmentation is a practical solution for extracting structured visual information from biomedical images^{20–29}, enabling the identification of regions of interest (ROI) across organs, lesions, tissues, and cells. Although recent segmentation foundation

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models^{20,22,30,31} have demonstrated remarkable performance in this task, they cannot generally generate diagnostic findings (e.g., clinical reports), limiting their practicality in AI-assisted biomedical image analysis.

To address these challenges, we highlight the importance of grounded interpretation in biomedical image analysis, *i.e.*, generating diagnostic findings and simultaneously segmenting the corresponding biomedical targets. In this way, we enable biomedical AI models to deliver both accurate and interpretable results for effectively assisting clinicians in biomedical image analysis. In this work, we introduce UniBiomed, the first universal foundation model for grounded biomedical image interpretation. UniBiomed innovatively integrates advanced Multi-modal Large Language Models (MLLMs)^{32,33} and Segment Anything Model^{34,35} (SAM) for grounded biomedical image interpretation, as shown in Fig. 1c. Specifically, the MLLM is used to interpret multi-modal biomedical images and generate diagnostic findings. Then, we combine the encoded user instructions and the outputs of MLLM to prompt the SAM model for segmenting the biomedical objects corresponding to the generated findings. Through this integration, UniBiomed can tackle a wide range of biomedical tasks, including biomedical image segmentation, disease recognition, region-aware diagnosis, vision question answering (VQA), and report generation. These advancements enable UniBiomed to simultaneously provide fine-grained visual and textual information for fine-grained biomedical image analysis.

The power of UniBiomed originates from its ability to leverage holistic multi-modal biomedical information in universal training. Concretely, previous biomedical models typically rely on disjoint training with independent datasets, e.g., clinical report datasets for report generation training^{15,19,36,37}, segmentation datasets for segmentation training^{20,22,25}. In contrast, our UniBiomed is versatile in utilizing different types of biomedical datasets to improve the performance of biomedical image analysis complementarily. Within a universal training process, training with segmentation datasets enables UniBiomed to extract critical biomedical objects, region-aware diagnosis datasets enhance the region-aware ability, and VQA and report generation datasets improve the model's capability to interpret clinicians' instructions and produce accurate diagnostic findings. Integrating complementary multi-modal datasets for training, UniBiomed demonstrates more powerful and versatile abilities in biomedical image analysis.

To develop UniBiomed, as shown in Fig. 1a, b, we curate a large-scale dataset containing 27 million triplets of images, region annotations, and text descriptions spanning ten biomedical imaging modalities. Region annotations include both segmentation masks and bounding boxes, providing detailed spatial localization information for the model to develop region-aware capabilities. The text descriptions are extracted from readily available clinical texts accompanying public datasets, encompassing biomedical definitions, diagnostic findings, medical knowledge, and clinical reports. To the best of our knowledge, this is the largest and most comprehensive dataset for biomedical grounded interpretation. By integrating comprehensive and multi-granular visual and textual biomedical information for training, UniBiomed unleashes the untapped grounded interpretation capability in biomedical image analysis and achieves superior performance across diverse biomedical tasks.

We conduct a large-scale validation on 70 internal and 14 external datasets across 10 diverse biomedical imaging modalities. Extensive experiments demonstrate the effectiveness of UniBiomed. Specifically, in biomedical image segmentation, UniBiomed surpasses state-of-the-art segmentation foundation models^{20,22,30,31} by a substantial margin, e.g., surpassing the representative model BiomedParse²² by an average of 10.25% in Dice scores across 60 segmentation datasets. Beyond segmentation, UniBiomed also achieves superior performance in diverse biomedical tasks compared with state-of-the-art biomedical

MLLMs, including disease recognition, VQA, ROI classification, region-aware report generation, and report generation. More importantly, we further showcase that UniBiomed is an effective biomedical AI tool for optimizing the biomedical image analysis workflow. Specifically, to recognize and localize abnormalities in images, previous biomedical AI models^{20,22,30,31} heavily rely on clinical experts to provide accurate textual or visual prompts, e.g., instruct the model that the input image contains tumors (text prompts) or provide tight bounding boxes to highlight the tumors in the images (visual prompts). In contrast, UniBiomed can eliminate these processes and enable automated end-to-end grounded interpretation of biomedical images, effectively streamlining the analysis workflow. In summary, UniBiomed represents a versatile and powerful foundation model that delivers superior performance in grounded biomedical image interpretation, demonstrating promising potential towards more accurate and efficient biomedical image analysis.

Results

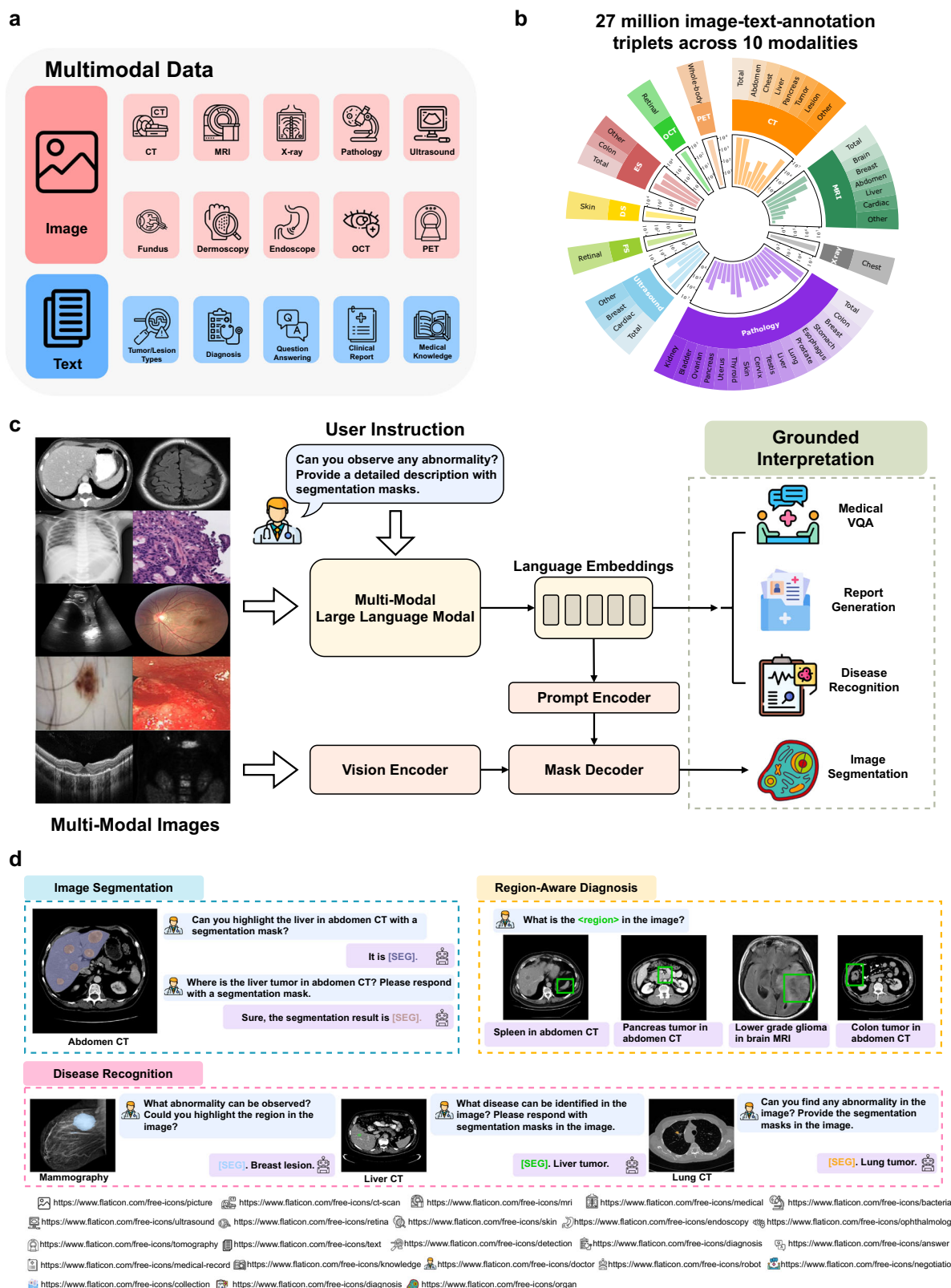
Overview of uniomed

Dataset. UniBiomed is designed to address the biomedical-grounded interpretation tasks, necessitating training datasets to provide both region information and expert-annotated text descriptions. For the region information, we collect the readily available segmentation masks and bounding boxes from public biomedical datasets. Specifically, the segmentation masks are used for dense segmentation training. The bounding boxes are used for ROI classification and region-aware report generation training. Each of these segmentation masks and bounding boxes is paired with a specific text description, providing the biomedical classes or diagnostic findings of the annotated biomedical objects. To ensure high-quality text descriptions, we extract semantic labels, diagnostic findings, medical knowledge, and clinical reports from publicly available datasets following previous methods^{22,36,37}. These descriptions capture fine-grained biomedical information across multiple granularities, including organs, lesions, tissues, and cells, enabling comprehensive interpretation of biomedical targets.

A critical step in our dataset pre-processing pipeline is transforming the textual descriptions into a uniform Vision-Question-Answering (VQA) format^{19,32,38,39}, aiming to facilitate universal training. For example, given a CT image with liver tumors, we define the question as: "Can you identify any abnormality within this CT image? Please respond with segmentation masks.". Then the corresponding answer is structured as: "It is [SEG]. Liver tumor". For the ROI classification task without segmentation outputs, the answer will only contain the biomedical class without the "[SEG]" token^{38,39}. This uniform VQA format enables UniBiomed to jointly recognize abnormalities and segment biomedical targets within a universal foundation model. Some VQA examples are shown in Supplementary Fig. 4.

Through this process, we construct a large-scale dataset consisting of 27 million image-text-annotation triplets across 10 modalities, as shown in Fig. 1a, b. Specifically, the 3D medical images (CT and MRI) are pre-processed as 2D slices following previous methods^{20,22}. To the best of our knowledge, this is the largest and most comprehensive dataset for biomedical grounded interpretation. For example, compared with the dataset used in the representative biomedical foundation model BiomedParse²², our dataset is over 30 times larger in scale. Unleashing the power of this large-scale dataset, UniBiomed demonstrates state-of-the-art performance across diverse tasks, outperforming existing biomedical foundation models^{19,20,22,30,31,40,41} by a large margin. The details of the used datasets are presented in Supplementary Tables 2–5.

Method. Multi-modal large language models (MLLMs) have achieved remarkable effectiveness in processing vision-language information^{32,42–45}. Typically, MLLMs leverage a vision encoder to



encode images as vision tokens and input them into a Large Language Model (LLMs)⁴⁶ for text generation³². However, state-of-the-art MLLMs^{32,42–44} still fail to output the localizations of the corresponding regions described in the generated texts. This task, known as grounded interpretation in computer vision^{38,39,47–50}, requires a strong segmentation model to segment the target regions. SAM^{34,35} has demonstrated promising results in segmenting diverse sources of

images, including biomedical images^{20,22}. The advances in MLLMs^{32,42–44} and SAM^{34,35} lead to a promising way for us to investigate grounded biomedical image interpretation.

In this work, we introduce UniBiomed, which innovatively integrates MLLM^{32,33} and SAM^{34,35} for universal grounded interpretation of biomedical images. The overall framework is shown in Fig. 1c. Our approach combines the complementary strengths of MLLM and SAM:

Fig. 1 | Overview of the study. **a** UniBiomed enables universal grounded interpretation across 10 different biomedical imaging modalities. CT, computed tomography; MRI, magnetic resonance imaging; OCT, optical coherence tomography; PET, positron emission tomography. **b** We curated 27 million triplets of images, region annotations (segmentation masks and bounding boxes), and text descriptions for training UniBiomed. The text descriptions are processed from readily available diagnostic findings in public datasets. **c** The framework of UniBiomed. UniBiomed incorporates Multi-modal Large Language Models (MLLMs)^{32,33} with Segment Anything Model^{34,35} (SAM) to tackle text description and segmentation tasks jointly. MLLM contains both vision and language models to interpret visual and language information, ultimately generating clinical text descriptions. SAM includes a vision encoder, a prompt encoder, and a mask decoder to address the segmentation task. Given user instructions with biomedical images, UniBiomed can segment target objects and generate grounded text descriptions simultaneously, enabling end-to-end analysis of multi-modal biomedical images. **d** UniBiomed is designed to unify diverse biomedical tasks within a universal training process, including segmentation, disease recognition, region-aware diagnosis, vision question answering (VQA), and report generation. In (c), the used

images are from refs. 11,86–93. In (d), the images used in “Image Segmentation” are from ref. 11, the images used in “Region-Aware-Diagnosis” are from⁴¹, the images used in “Disease Recognition” are from refs. 11,94. Icon made by Pixel perfect from www.flaticon.com/free-icons/picture, <https://www.flaticon.com/free-icons/ct-scan>, <https://www.flaticon.com/free-icons/mri>, <https://www.flaticon.com/free-icons/medical>, <https://www.flaticon.com/free-icons/bacteria>, <https://www.flaticon.com/free-icons/ultrasound>, <https://www.flaticon.com/free-icons/retina>, <https://www.flaticon.com/free-icons/skin>, <https://www.flaticon.com/free-icons/endoscopy>, <https://www.flaticon.com/free-icons/ophthalmology>, <https://www.flaticon.com/free-icons/tomography>, <https://www.flaticon.com/free-icons/text>, <https://www.flaticon.com/free-icons/detection>, <https://www.flaticon.com/free-icons/diagnosis>, <https://www.flaticon.com/free-icons/answer>, <https://www.flaticon.com/free-icons/medical-record>, <https://www.flaticon.com/free-icons/knowledge>, <https://www.flaticon.com/free-icons/doctor>, <https://www.flaticon.com/free-icons/robot>, <https://www.flaticon.com/free-icons/negotiation>, <https://www.flaticon.com/free-icons/collection>, <https://www.flaticon.com/free-icons/organ>.

MLLM processes both visual and textual information to generate descriptive interpretations, while SAM performs precise image segmentation through its vision encoder, prompt encoder, and mask decoder modules.

The framework operates as an end-to-end way that, given user instructions and biomedical images, UniBiomed simultaneously: (1) generates text descriptions of the biomedical images, and (2) segments the corresponding biomedical targets. The effectiveness of uniomed arises from the complementary integration of MLLM and SAM. Specifically, UniBiomed combines the output of the MLLM^{38,39,51} with the tokenized user instructions as the language embeddings to guide SAM's segmentation process. This incorporation effectively eliminates the manual efforts of crafting bounding boxes^{20,30} or detailed text descriptions^{31,52} as prompts for segmentation. For implementation, we select InternVL2.5^{33,43} as our foundation MLLM due to its robust multi-modal understanding capabilities, and SAM2³⁵ is adopted as our segmentation model for its improved segmentation performance. The details of the method are described in Section 4.1.

Evaluation. To demonstrate the effectiveness of UniBiomed on diverse biomedical tasks, we conduct a comprehensive evaluation on 70 internal and 14 external datasets. Since we focus on evaluating the effectiveness of grounded interpretation, for comparison methods, we compare segmentation foundation models^{20,22,30,31} on segmentation tasks, compare grounding tasks with grounding methods^{39,51}, and compare medical MLLMs^{19,40,41} on medical diagnosis tasks. (1) First, we conduct extensive comparisons with several foundation models^{20,22,30,31} in biomedical image segmentation. (2) Second, we evaluate UniBiomed in a challenging grounded VQA task, i.e., grounded disease recognition, which aims to simultaneously generate diagnostic findings and segment the corresponding biomedical targets. (3) Then, we conduct extensive experiments on grounded report generation, which is designed to generate clinical reports of the given images and highlight the target regions. We verify the grounded report generation task on the RadGenome^{37,53,54} dataset. (4) Finally, we verify the effectiveness of UniBiomed in two region-aware diagnosis tasks: ROI classification and region-aware report generation. Specifically, we extract bounding boxes from segmentation masks to indicate the ROIs following previous methods^{41,55–57}. We assess the region-aware report generation task using the MedTrinity³⁶ dataset. The details of the datasets are presented in Supplementary Tables 2–5.

UniBiomed excels in biomedical image segmentation

In this section, we present a comprehensive comparison with multiple biomedical segmentation foundation models in biomedical image segmentation, i.e., MedSAM²⁰, SegVol³⁰, SAT³¹, and BiomedParse²², as

shown in Fig. 2. Specifically, we evaluate our method on 46 internal and 14 external datasets across nine diverse biomedical imaging modalities. The details of the datasets are shown in Supplementary Table 2. We adopt text prompts for referring segmentation as SAT³¹ and BiomedParse²². For example, to segment liver tumors in the CT image, we provide a text instruction “Please segment liver tumor in the CT image” to the model, eliminating the need to offer visual prompts like bounding boxes or points as MedSAM²⁰ and SegVol³⁰. More details of the competing methods are presented in Section 4.

As shown in Fig. 2a, b, UniBiomed achieves superior performance compared to state-of-the-art segmentation foundation models^{20,22,30,31}. Specifically, UniBiomed surpasses the best-competing method BiomedParse²² by a clear margin, i.e., 10.25% dice score improvements on 60 datasets across nine modalities. Figure 2d shows that UniBiomed outperforms BiomedParse by 9.13% and 13.95% in 46 internal and 14 external datasets, respectively. These findings robustly underscore UniBiomed's substantial advancement in universal biomedical image segmentation.

One of the key reasons for UniBiomed's breakthrough in biomedical image segmentation lies in its ability to unleash the power of MLLM and SAM in large-scale multi-modal biomedical datasets. Unlike prior segmentation models^{20,22,30,31} that rely solely on segmentation datasets for training, UniBiomed effectively incorporates large-scale VQA and report generation datasets^{36,37} in universal training. This approach enables UniBiomed to develop more generalizable representations, leading to superior segmentation performance across diverse biomedical imaging modalities. As shown in Fig. 2e, with large-scale multi-modal datasets for training, we improve UniBiomed by 3.39% and 8.26% dice scores in internal and external validation, respectively.

More importantly, beyond segmentation, UniBiomed establishes a more comprehensive interpretation capability for biomedical image analysis, unlocking novel opportunities in broader clinical applications. As illustrated in Fig. 2f, prior segmentation models^{20,22,30,31} are still limited in tackling the segmentation tasks, which heavily hamper their clinical applications in biomedical image analysis. In contrast, UniBiomed extends segmentation to grounded interpretation with text generation, which is a versatile foundation model capable of handling diverse biomedical tasks, making it more adaptable for clinical applications.

UniBiomed enables accurate grounded disease recognition

In medical MLLMs^{15,16,19,37,52,58,59}, disease recognition is one of the most important VQA tasks, which aims to identify the lesions, tumors, or abnormalities in the given images. However, existing medical MLLMs^{15,16,19,37,52,58,59} are typically region-agnostic, which solely generate

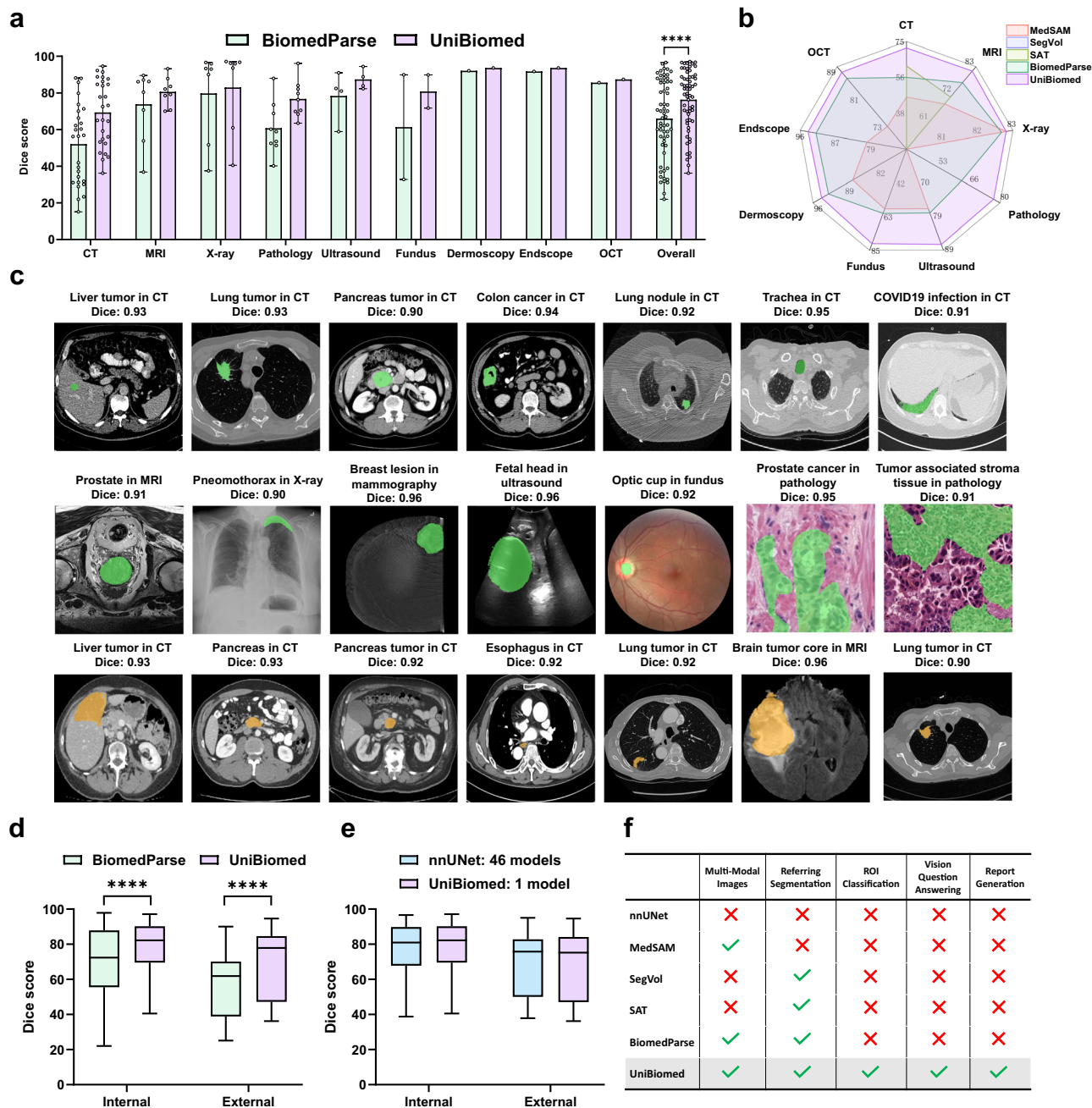


Fig. 2 | Comparison of biomedical image segmentation. **a** Comparison of Dice scores between UniBiomed and the best-competing segmentation foundation model BiomedParse²² across nine modalities, with the mean ($\pm$ s.d.) reported. Significance levels at which UniBiomed outperforms BiomedParse²², with a two-sided paired t-test P -value of **** $P < 1 \times 10^{-4}$. Exact P -value is 6.048×10^{-6} . Notably, UniBiomed outperforms BiomedParse²² by an average of 10.25% in dice scores on 60 internal and external datasets. The statistics are obtained from five times of experiments. **b** Radar chart comparisons with several representative segmentation foundation models, MedSAM²⁰, SegVol³⁰, SAT³¹, and BiomedParse²². Specifically, SAT³¹ is only applicable to 3D modalities like CT and MRI. SegVol³⁰ is solely available for CT datasets. **c** Qualitative segmentation results of UniBiomed. Specifically, the colors of segmentation masks for internal and external validations are green and orange, respectively. More visualization results are presented in Supplementary Fig. 5. **d** Box plot comparisons between UniBiomed and BiomedParse²². Significance levels at which UniBiomed outperforms BiomedParse²², with two-sided paired t-test P -values of **** $P < 1 \times 10^{-4}$ for both internal and external validations. Exact P -values are 4.017×10^{-5} and 1.043×10^{-6} , respectively. The statistics are obtained from five times of experiments. Box plots show the mean (center), 25 and 75th percentiles

(bounds of box), and minima to maxima (whiskers). **e** Ablation studies of UniBiomed. We compare UniBiomed under two settings: (1) train on only segmentation datasets, and (2) train on large-scale multi-modal datasets, including segmentation, VQA, and report generation datasets. It can be seen that with large-scale multi-modal datasets for training, the segmentation performance can be further improved, the two-sided paired t-test P -values are ** $P < 1 \times 10^{-2}$ and **** $P < 1 \times 10^{-4}$ for internal and external validations, respectively. Exact P -values are 8.29×10^{-3} and 2.38×10^{-5} , respectively. The statistics are obtained from five times of experiments. Box plots show the mean (center), 25 and 75th percentiles (bounds of box), and minima to maxima (whiskers). **f** Overall application comparisons with representative biomedical segmentation models^{20,22,30,31}. Referring segmentation represents using text instructions as prompts for segmentation, which eliminates the efforts of providing tight bounding boxes for the biomedical objects²⁰. Notably, UniBiomed not only supports multi-modal biomedical image segmentation but also establishes remarkable capabilities in ROI classification, VQA, and report generation, where state-of-the-art segmentation models^{20,22,30,31} fail to establish. In c, the used images are from refs. 11,86–89,95–97.

diagnostic findings but fail to localize relevant regions (e.g., tiny lesions) in the images. This lack of spatial awareness severely limits their clinical applications, as precise localization of abnormalities is essential for diagnosis and treatment planning.

In this work, we introduce a challenging grounded VQA task, *i.e.*, grounded disease recognition, which is designed to simultaneously generate diagnostic findings and segment the corresponding targets. In this task, the model is asked to predict abnormality class (e.g., liver tumor or pancreas tumor) and segment the targets in the images. If there is no abnormality in the given images, the model is supposed to output “No findings” and produce all-zero segmentation masks.

To this end, we curate a comprehensive dataset encompassing 15 distinct abnormality types for model training and validation, as shown in Fig. 3a. Specifically, these 15 abnormality types contain liver tumor, pancreas tumor, kidney tumor, colon cancer, lung tumor, lung nodule, COVID-19 infection, fibrotic lung disease, brain tumor, breast lesion, colon polyp, pneumothorax, prostate cancer, skin lesion, and retinal lesion. The details of the datasets are shown in Supplementary Tables 2 and 6.

Existing medical models lack the capability to tackle this task, which requires simultaneously outputting segmentation masks and diagnostic findings. Although BiomedParse²² can conduct segmentation and employ a meta-object classifier for classification, this classifier requires users to pre-diagnose the images and provide diagnostic findings as textual prompts to the model, thus failing to perform the grounded disease recognition task introduced in this work. To this end, we re-implement two state-of-the-art approaches from the general domain, LISA³⁹ and GLaMM⁵¹, on our curated datasets for fair comparisons. The comparison results are shown in Fig. 3b–d. Specifically, UniBiomed outperforms the best competing method LISA³⁹ by 3.86% and 3.29% in segmentation dice scores and disease recognition accuracy, respectively. Consistent improvements across different types of abnormalities are observed in Fig. 3d, which robustly validates the effectiveness of UniBiomed in this challenging VQA task.

More importantly, the strong capability of UniBiomed in grounded disease recognition potentially leads to a significant paradigm shift in the biomedical image analysis workflow. Specifically, previous biomedical foundation models heavily rely on clinical experts to pre-diagnose images and manually craft precise textual or visual prompts. For example, for the task of lung tumor recognition, one of the representative foundation models, MedSAM²⁰, requires radiologists to outline the regions of lung tumors in each slice of a 3D CT scan and provide the prompts to the model. Another representative foundation model, BiomedParse²², requires radiologists to identify lung tumors in each slice of a 3D CT scan in advance, then input an instruction “Please segment lung tumor in this CT image” to prompt the model for lung tumor segmentation. In contrast, UniBiomed enables an end-to-end pipeline to recognize abnormalities in images. As shown in Fig. 3e, UniBiomed is free of pre-diagnosis by clinical experts. Instead, UniBiomed adopts a template instruction and automatically recognizes the abnormality class with segmentation predictions. These findings underscore the clinical practice of UniBiomed’s grounded interpretation capabilities, which is a non-trivial advancement compared to state-of-the-art biomedical foundation models^{20,22,30,31}.

The clinical impact of dice scores improvements. We would like to discuss the clinical impact of dice scores improvements, mainly in two aspects: (1) Higher dice scores generally mean better abnormality detection, which can potentially improve the survival rate of patients. For example, previous works^{60,61} generally employ segmentation models to detect cancers in CT scans, where higher dice scores generally indicate better detection accuracy. An example in Supplementary Fig. 5b can be used to illustrate this point, e.g., UniBiomed correctly detects colon cancer in the image, which results in higher dice scores. (2) Higher dice scores generally mean more precise size

and localization measurement, which can potentially improve the diagnosis and decision-making. Take lung nodule diagnosis as an example, the precise measurement of size growth is critical for its risk stratification⁶², and a model with higher dice scores is essential to this task.

UniBiomed enables accurate grounded report generation

Grounded report generation is a challenging task that combines biomedical image segmentation with report generation, enabling end-to-end biomedical image analysis in a universal model. Unlike prior methods^{19,22,40,41}, which rely on separate segmentation models and standalone language models for report generation, UniBiomed performs both tasks within one universal model. This integration allows UniBiomed to leverage holistic biomedical knowledge more effectively, improving both segmentation and report quality.

To evaluate its effectiveness, we conduct extensive comparisons on the RadGenome^{53,54} dataset. Specifically, the RadGenome dataset is processed from the CT-RATE³⁷ dataset, which contains detailed CT reports with segmentation masks for training and validation. The clinical reports provide detailed descriptions of the images, and the segmentation masks indicate the localization of corresponding organs. Since existing medical MLLMs^{15,16,19,40,41,52,58,59} cannot tackle this task, we also re-implement GLaMM⁵¹ and LISA³⁹ on our datasets for comparisons. The results are shown in Fig. 3f, g.

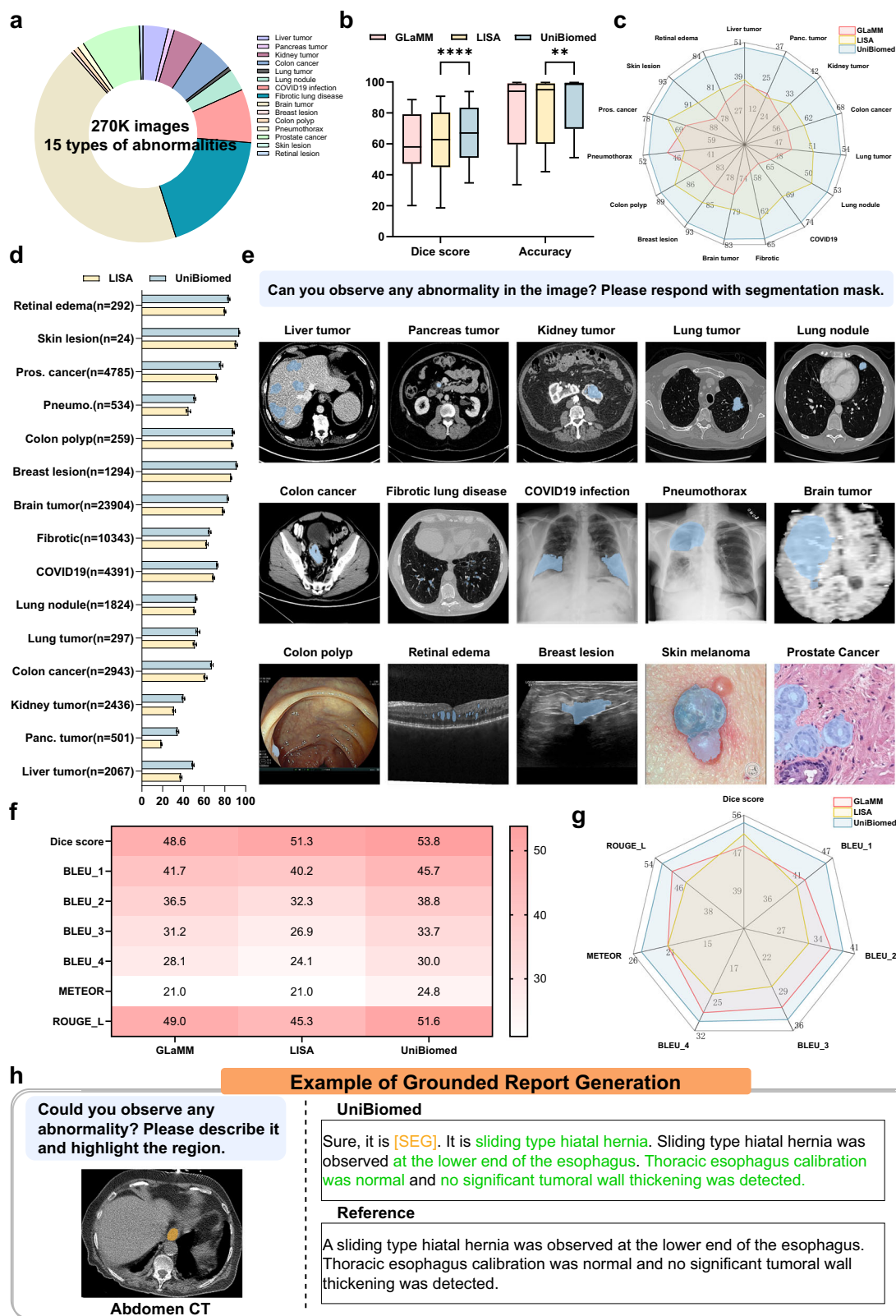
We report the dice scores results for segmentation evaluation, Bilingual Evaluation Understudy (BLEU)⁶³, Metric for Evaluation of Translation with Explicit Ordering (METEOR)⁶⁴, and Recall-Oriented Understudy for Gisting Evaluation (ROUGE_L)⁶⁵ results for report generation evaluation. It can be observed that UniBiomed not only achieves better segmentation performance (53.8% dice scores) but also demonstrates superior report generation capabilities (45.7%, 24.8%, and 51.6% in BLEU_1, METEOR, and ROUGE_L, respectively).

UniBiomed effectively improves region-aware diagnosis

Region-aware diagnosis refers to recognizing the biomedical objects within the user-defined ROIs in the images^{36,41}. We involve this task in our training to enhance the region-aware ability of UniBiomed and ultimately improve the performance of grounded interpretation. Following the settings of MedPLIB⁴¹, we evaluate two region-aware diagnosis tasks in this work, *i.e.*, ROI classification and region-aware report generation. Since the segmentation predictions are not required in this task, we output the results by the MLLM module only. Following previous methods^{36,56}, to indicate the ROIs, we directly overlay bounding boxes onto the images, eliminating the need for complex region encodings for indicating ROIs. The details are presented in Section 4.

We first evaluate the performance of ROI classification. For fair comparisons, we compare UniBiomed with two state-of-the-art medical MLLMs, MedRegA⁴⁰ and MedPLIB⁴¹, since both of them were trained on large-scale ROI classification datasets. The results are shown in Fig. 4a–d. Specifically, UniBiomed achieves an average 93.38% accuracy in ROI classification, surpassing the best competing method MedPLIB⁴¹, by an average of 8.32% in ROI classification accuracy across ten imaging modalities. These results validate that UniBiomed also excels in ROI classification compared with state-of-the-art methods.

We further evaluate a more challenging region-aware diagnosis task, *i.e.*, region-aware report generation, which aims to generate detailed reports within ROIs in the biomedical images. Our experiments are conducted on the MedTrinity³⁶ dataset, a large-scale benchmark featuring region-centric reports across diverse biomedical imaging modalities. Specifically, MedTrinity is aggregated from 23 public VQA and report generation datasets across ten biomedical imaging modalities, as described in Supplementary Table 3. We compare our approach with several state-of-the-art MLLMs, including InternVL2.5^{33,43}, LLaVA-Med¹⁹, MedRegA⁴⁰, and MedPLIB⁴¹. Among them, InternVL2.5 is the state-of-the-art MLLM in the general domain.



LLaVA-Med, MedRegA, and MedPLIB are generalist MLLMs in the medical domain, while MedRegA and MedPLIB were trained on large-scale region-centric report generation datasets. We report the results of BLEU⁶³, METEOR⁶⁴, and ROUGE⁶⁵.

The results are shown in Fig. 4e, f. It can be seen that UniBiomed achieves superior performance compared with previous MLLMs^{19,33,40,41}. Specifically, UniBiomed achieves 52.4%, 30.4%, and

47.9% in BLEU_1, METEOR, and ROUGE_L, respectively, surpassing the best-competing method MedPLIB⁴¹ by 8.8, 2.3, and 15.8%. These results robustly validate the effectiveness of UniBiomed in this task. We further present an example of region-aware report generation on prostate pathology images, as shown in Fig. 4g. It can be seen that UniBiomed can generate a detailed description to illustrate the observed features. Specifically, UniBiomed not only accurately identifies the abnormality

Fig. 3 | Comparison of grounded disease recognition and grounded report generation. **a** We curate a new dataset for grounded disease recognition. Specifically, this dataset contains 270K images with 15 types of abnormalities. The details of the dataset are presented in Supplementary Table 6. **b** We compare UniBiomed with two state-of-the-art methods, LISA³⁹ and GLaMM⁵¹, in segmentation dice scores and disease recognition accuracy. Notably, UniBiomed outperforms the best competing method LISA³⁹ by 3.86% in dice scores and 3.29% in accuracy, with two-sided paired t-test P -values of $****P < 1 \times 10^{-4}$ and $**P < 1 \times 10^{-2}$. Exact P -values are 1.85×10^{-5} and 2.81×10^{-3} , respectively. The statistics are obtained from five times of experiments. Box plots show the mean (center), 25th and 75th percentiles (bounds of box), and minima to maxima (whiskers). **c** Radar chart comparisons across 15 types of abnormalities. **d** Dice scores comparisons between UniBiomed and LISA³⁹

across 15 types of abnormalities, with the mean ($\pm$ s.d.) reported. The statistics are obtained from five times of experiments. **e** The qualitative visualization results of grounded disease recognition. The template user instruction is shown in the blue box. Given template instructions, UniBiomed can recognize the abnormalities in the images and highlight the precise localizations. More visualization results are presented in Supplementary Fig. 5. **f** Comparisons of grounded report generation on the RadGenome^{53,54} dataset. We re-implement GLaMM⁵¹ and LISA³⁹ on this dataset for comparison. We report dice scores results for segmentation evaluation, BLEU⁶³, METEOR⁶⁴, and ROUGE_L⁶⁵ results for report generation evaluation. **g** Radar chart comparisons across different metrics in grounded report generation. In (e), the used images are from^{11,86,87,95,96}.

(i.e., pneumothorax) within the target regions, but also generates a detailed description to illustrate the observed features. More examples are presented in Supplementary Fig. 6.

UniBiomed optimizes biomedical image analysis workflow

In this section, we show that UniBiomed is an effective AI tool for improving the efficiency of the biomedical image analysis workflow. As shown in Fig. 5a, previous biomedical segmentation foundation models, such as MedSAM²⁰ and BiomedParse²² follow a cumbersome process. Given an input CT scan, these models require radiologists to first pre-diagnose each slice to locate target abnormalities, such as liver disease in the example. Once identified, additional manual inputs are necessary, e.g., BiomedParse²² demands a text description as the textual prompt, while MedSAM²⁰ relies on a tightly drawn bounding box prompt. More critically, these models are limited to segmentation tasks and fail to generate detailed diagnostic descriptions of the targets. These inefficiencies significantly hinder their clinical applications.

In contrast, UniBiomed overcomes these limitations through an integration of MLLM and SAM. Given an input CT scan, UniBiomed automatically detects abnormalities, generates precise segmentation masks, and provides detailed diagnostic descriptions. There are two significant advantages in this workflow: (1) First, this process is free of pre-diagnosis by radiologists and requires no manual prompts, eliminating the time-consuming process of slice-by-slice analysis and manual prompt engineering. (2) Moreover, unlike prior foundation models^{20,22,30,31}, which are limited to segmentation outputs, UniBiomed is capable of combining visual analysis with diagnostic text generation. For example, UniBiomed not only segments targets but also delivers clinically relevant interpretations. These advancements lead to a significant paradigm shift in biomedical image analysis, which optimizes workflow efficiency and enables an end-to-end diagnosis workflow.

In addition, for 3D medical image analysis, our grounded interpretation does not require pre-diagnosis and slice selection. We use an example of lung tumor detection on 3D CT images to illustrate this advancement, as shown in Supplementary Fig. 8. Given a 3D volume for abnormality detection, UniBiomed does not demand radiologists to diagnose each slice and provide visual prompts like bounding boxes²⁰. Instead, we use a template instruction (e.g., “What abnormality can be seen in this image? Please respond with the segmentation mask”) to prompt the UniBiomed, simultaneously delivering diagnostic results and segmentation masks for all of the slices. In contrast, prior models^{20,22} require radiologists to pre-diagnose and select the slices.

We further conduct a reader study on 30 CT scans, including 10 lung tumor cases, 10 liver tumor cases, and 10 pancreas tumor cases, as shown in Supplementary Table 24. One radiologist is involved in comparing our model with two competing models, MedSAM²⁰ and BiomedParse²². Our reader study shows that UniBiomed is more efficient than previous models^{20,22}, which is preferred by the radiologist.

Discussion

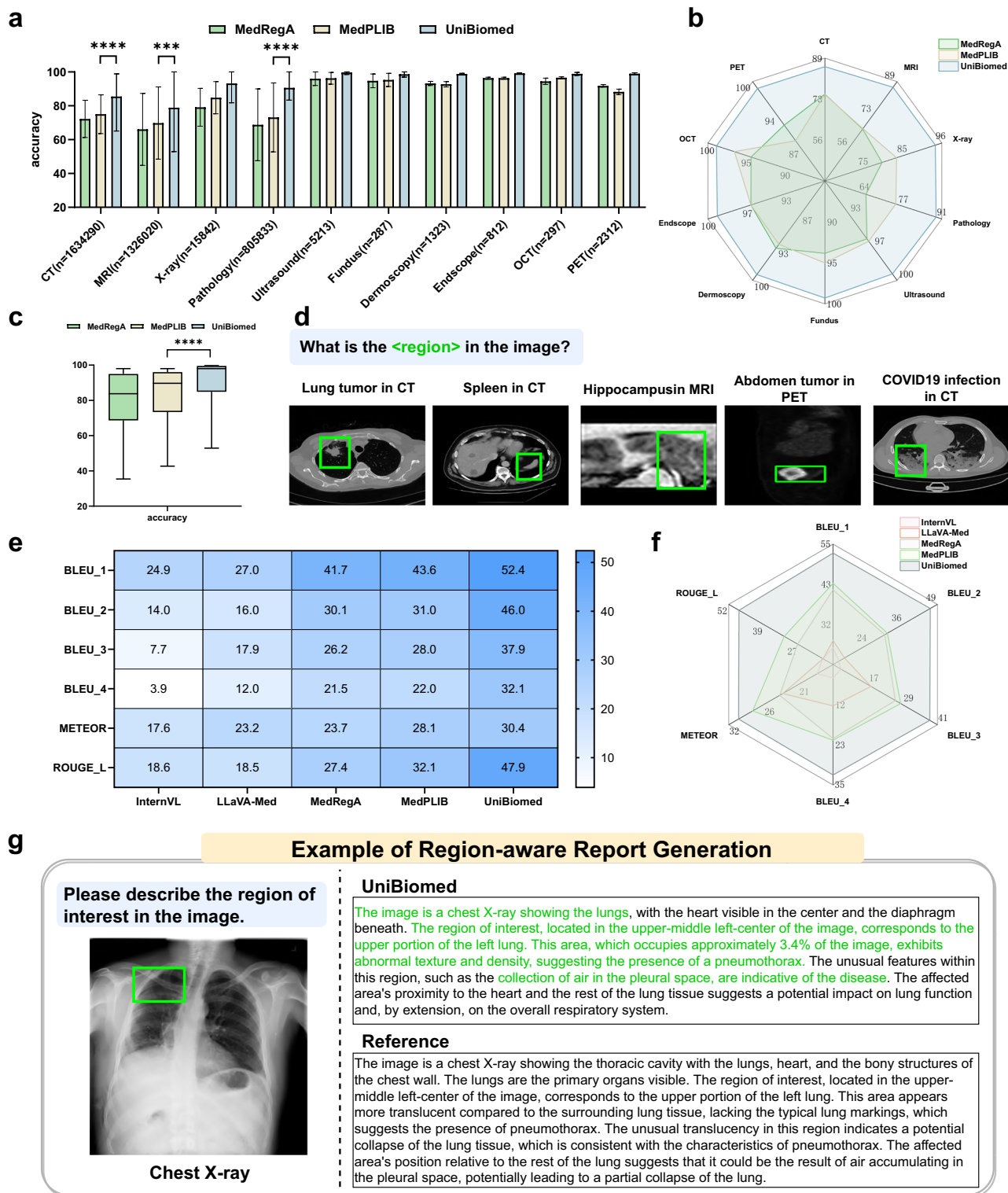
In this work, we introduce UniBiomed, the first universal foundation model for grounded biomedical image interpretation. UniBiomed

effectively unifies the generation of diagnostic findings with the segmentation of corresponding biomedical targets, enabling accurate biomedical image analysis across ten diverse imaging modalities. UniBiomed is based on an integration of an advanced Multi-modal Large Language Model (MLLM) and SAM^{34,35}, which can effectively leverage multi-modal biomedical information for tackling diverse biomedical tasks. To develop UniBiomed, we curate a large-scale dataset comprising 27 million triplets of images, region annotations, and text descriptions spanning ten biomedical imaging modalities, which is the largest and most comprehensive dataset for biomedical grounded interpretation. To evaluate the effectiveness of UniBiomed, we conduct a large-scale validation on 84 datasets with comprehensive comparisons of state-of-the-art methods^{19,20,22,30,31,33,39–41,51}. To evaluate the generalizability of our model, we involve 14 external datasets in validation, which were unseen in the training. Extensive experiments demonstrate that UniBiomed achieves state-of-the-art performance on a wide range of biomedical tasks, including biomedical image segmentation, disease recognition, region-aware diagnosis, and report generation. We further showcase that UniBiomed represents a paradigm shift in biomedical image analysis workflows, which effectively improves the efficiency of AI-assisted biomedical image analysis. These findings demonstrate the promising prospects of UniBiomed in clinical applications.

Recent advances have attracted increasing attention to multi-modal information for biomedical image analysis^{15,16,19,52,58,59,66,67}. Concretely, visual features extracted from biomedical images provide anatomical and functional information from cell to organ levels^{9–11,21,22,68–70}, while textual descriptions from clinical experts offer fine-grained information for interpreting biomedical images^{12,13,15,71}. Although recent biomedical AI models have showcased encouraging results in biomedical image analysis, it remains challenging to effectively integrate holistic vision and language information for assisting clinicians in practice. Specifically, the clinical adoption of biomedical AI models requires that the AI-generated results are both accurate and interpretable for clinicians to ensure reliability. Thus, it demands that biomedical AI models can output the diagnostic findings and simultaneously highlight the corresponding regions in the images, offering rich visual and textual information for assisting clinicians.

To this end, we underscore the importance of grounded interpretation in biomedical image analysis, i.e., generating diagnostic findings and simultaneously segmenting the corresponding biomedical objects. This task enables models to extract important biomedical targets (e.g., lesions, tumors, cancer cells, or abnormal organs) and generate detailed descriptions (e.g., clinical reports) of corresponding objects to assist clinical experts in diagnosis, unlocking novel opportunities for end-to-end biomedical image analysis.

However, existing biomedical models^{19–22,30,31,39–41,51,52,58,59,72} fail to tackle this challenging yet critical task. In particular, grounded interpretation requires simultaneous execution of segmentation and text generation, whereas current models can only conduct these two tasks independently. Concretely, the biomedical segmentation foundation models^{20,22,30,31,72} can deliver segmentation masks while failing to



establish text generation tasks, such as VQA and report generation. Although recent medical MLLMs^{16,19,40,41,52,58,59} have demonstrated promising results in generating text descriptions for biomedical images, these models are not capable of segmenting the corresponding biomedical objects (e.g., tiny lesions) simultaneously, making it difficult for clinicians to identify the corresponding regions described in the reports. To illustrate the drawbacks of these methods more clearly, we present a detailed comparison in Supplementary Table 1. Compared with previous methods, UniBiomed is the first foundation model to support multi-modal image interpretation, flexible user prompts,

segmentation prediction, text description generation, region-aware diagnosis, pixel-level grounding, and end-to-end training in one universal model. This breakthrough paves a promising path towards more accurate and efficient biomedical image analysis.

The ability to leverage different types of biomedical datasets is pivotal to the success of UniBiomed. Previous biomedical foundation models are typically based on disjoint training on separate datasets, e.g., text generation datasets for language models^{15,19} or segmentation datasets for segmentation models^{20,22,25}. This limitation impedes their ability to leverage holistic vision and language information for

Fig. 4 | Comparison of ROI classification and region-aware report generation. **a** Comparisons with MedRegA⁴⁰ and MedPLIB⁴¹ in ROI classification, with the mean accuracy ($\pm$ s.d.) reported. Significance levels at which UniBiomed outperforms MedPLIB⁴¹, with two-sided paired t-test P -values are **** $P < 1 \times 10^{-4}$, *** $P < 1 \times 10^{-3}$, and **** $P < 1 \times 10^{-4}$ for CT, MRI, and pathology, respectively. Exact P -values are 1.06×10^{-5} , 9.11×10^{-5} , and 4.58×10^{-6} , respectively. The statistics are obtained from five times of experiments. **b** Radar chart comparison with MedRegA⁴⁰ and MedPLIB⁴¹ across ten diverse biomedical imaging modalities. **c** UniBiomed surpasses MedPLIB⁴¹ by 8.32% in ROI classification accuracy, with two-sided paired t-test P -values of **** $P < 1 \times 10^{-4}$. Exact P -value is 5.618×10^{-5} . The statistics are obtained from five times of experiments. Box plots show the mean (center), 25th and 75th percentiles (bounds of box), and minima to maxima (whiskers).

d Qualitative visualization results of ROI classification. The template user instruction is shown in the blue box. Specifically, given a bounding box prompt, UniBiomed can effectively predict the class of biomedical targets within ROIs. **e** Comparisons on region-aware report generation. We compare UniBiomed with multiple representative MLLMs, including InternVL2.5^{33,43}, LLaVA-Med¹⁹, MedRegA⁴⁰, and MedPLIB⁴¹. We report the results of BLEU⁶³, METEOR⁶⁴, and ROUGE_L⁶⁵. **f** Radar chart comparisons across different metrics in region-aware report generation. **g** An example of region-aware report generation on a chest x-ray image. The template user instruction is shown in the blue box. The text in green indicates the correct contents. In **(d)**, the used images are from refs. 11,96. In **(g)**, the used image is from ref. 86.

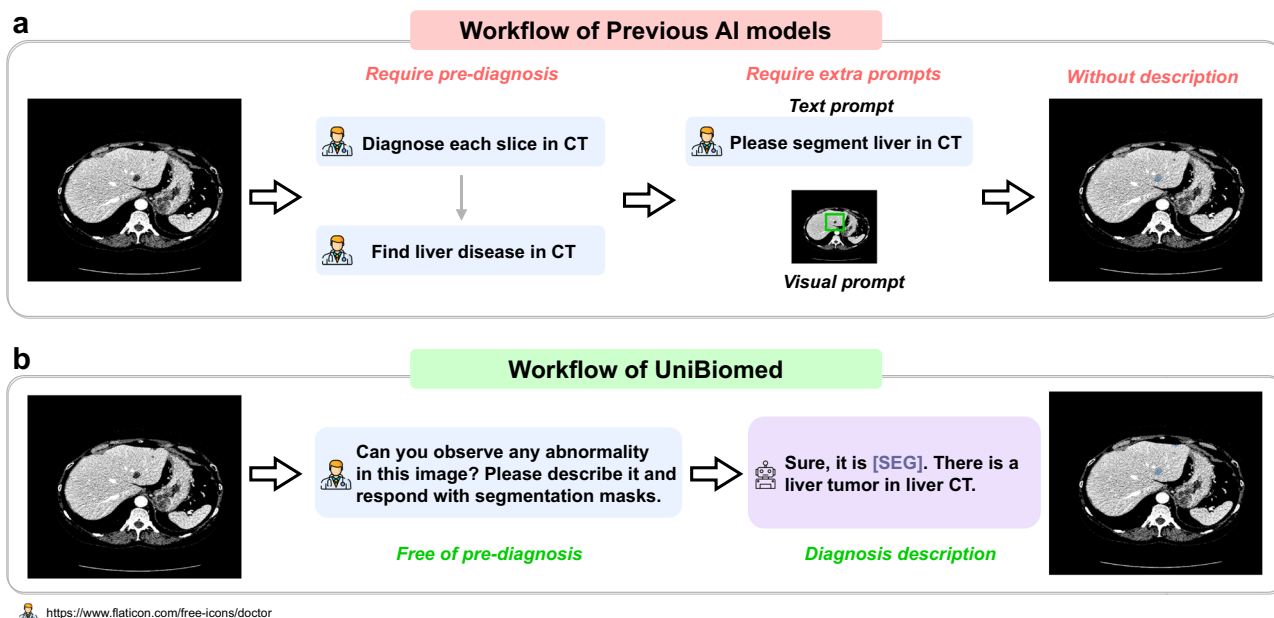


Fig. 5 | Workflow comparisons. **a** presents the workflow of previous biomedical segmentation foundation models, e.g., MedSAM²⁰ and BiomedParse²². **b** shows the workflow of our introduced UniBiomed. We present an example of grounded

report generation on CT images. Icon made by Pixel perfect from www.flaticon.com. The used images are from ref. 11. This is made by Freepik: <https://www.flaticon.com/free-icons/doctor>.

biomedical image analysis. In contrast, within a universal training process, UniBiomed can simultaneously leverage image segmentation datasets, vision question answering datasets, report generation datasets, and region-aware diagnosis datasets for developing a strong biomedical foundation model. This distinguished advantage makes UniBiomed stand out from previous biomedical foundation models, achieving better performance across diverse biomedical tasks. We present ablation studies to evaluate the effectiveness of universal training in Supplementary Fig. 3.

One of the major bottlenecks is the availability of data. To train UniBiomed, the training datasets necessitate both spatial localization and corresponding text descriptions for the biomedical objects. Inspired by the recent advances of “Visual Instruction Tuning” in MLLMs^{32,38,39}, we construct a uniform VQA format to facilitate universal grounded interpretation of biomedical images. In this way, we curate a large-scale dataset comprising 27 million image-text-annotation triplets across ten biomedical imaging modalities for training and validation, which is the largest and most comprehensive dataset in this field. Our curated dataset will fortify the foundation of future research in biomedical grounded interpretation.

We further showcase that UniBiomed is a more practical biomedical AI tool for optimizing workflow efficiency. Prior segmentation foundation models^{20,22,30,31} heavily rely on clinical experts to provide accurate textual or visual prompts. Take the representative models,

MedSAM²⁰ and BiomedParse²², as examples. Both of these models require the users to identify the targets of the input images in advance. Then, visual or textual prompts should be provided, e.g., MedSAM²⁰ requires tightly drawn bounding boxes as visual prompts while BiomedParse²² necessitates text instructions from pre-diagnostic findings as textual prompts. This process is tedious for clinicians and significantly hampers workflow efficiency. For example, for 3D medical images like CT and MRI, there are probably a few hundred slices within a scan, while these models^{20,22,30,31} require users to pre-diagnose the scans slice-by-slice and provide accurate prompts. In addition, these models^{20,22,30,31} still fail to generate diagnostic findings for interpreting the images, which significantly hinders their clinical applications. In contrast, UniBiomed provides automated end-to-end grounded interpretation of biomedical images, with both precise segmentation predictions and detailed text descriptions, which significantly optimizes the workflow efficiency of biomedical image analysis.

Although promising results have been demonstrated by UniBiomed, there are still several limitations for further improvement. Despite the extensive scale of our curated dataset, it may still lack incorporation of certain rare diseases or conditions, potentially limiting the model’s generalizability across all medical scenarios. For rare diseases, the bottleneck lies in the lack of annotated datasets for training and validation. We have added a rare disease dataset of adrenocortical carcinoma⁷³ in Supplementary Table 22. According to

the authoritative Orpha website (<https://www.orpha.net/>), adrenocortical carcinoma is a kind of rare disease. The results show that UniBiomed can also achieve competitive results on rare diseases if annotated datasets are available. This highlights the need for continuous dataset expansion and diversification to ensure comprehensive biomedical image analysis. In addition, while UniBiomed has achieved satisfactory performance on large-scale public datasets, further exploration of its clinical application is necessary to substantiate the effectiveness of our method. Moving forward, we will work closely with clinical practitioners to ensure that the model addresses practical needs and integrates seamlessly into existing workflows.

Discussion with medical agents

Medical agents^{74–76} have also received increasing attention in medical image analysis, which uses LLM as the orchestrator and calls different models to tackle various tasks. A key advantage of medical AI agents is their extensibility, which can be adapted for novel tasks by integrating new models. In contrast, universal foundation models^{15,19,22}, including UniBiomed, can tackle various tasks by only one model, which is more flexible and also eliminates the need to interoperate between different models. Recent work¹⁶ also demonstrated that a superior universal model can serve as a better orchestrator in medical agents. In the future, we will further investigate this valuable direction, exploring how to leverage the complementary strengths of medical agents and universal foundation models to advance medical image analysis.

Methods

Details of uniomed

UniBiomed is based on an integration of Multi-modal Large Language Model (MLLM)^{32,33} and the SAM^{34,35}, as shown in Fig. 6. Concretely, MLLM is responsible for interpreting multi-modal images and generating text descriptions, while SAM is employed to segment the target biomedical objects based on the given text instructions.

MLLM consists of a vision encoder⁷⁷ and a Large Language Model (LLM)⁴⁶. The vision encoder is a typical Vision Transformer (ViT)⁷⁷ model, which is pre-trained by CLIP⁴² for vision-language alignment^{32,43,44}. The LLM⁴⁶ is responsible for parsing the text instructions from users and generating corresponding answers. Given an image and a text instruction, the vision encoder encodes the image into visual tokens, while a text tokenizer⁷⁸ will process the text into language tokens. These visual and language tokens are then fed into the LLM to generate textual outputs, enabling UniBiomed to tackle diverse text generation tasks.

SAM^{34,35} is further leveraged for tackling the segmentation task, which comprises a vision encoder, a prompt encoder, and a mask

decoder. The vision encoder is based on Hiera⁷⁹, a hierarchical ViT⁷⁷ architecture. The image features extracted by SAM's vision encoder are passed to the mask decoder, where the prompt encoder encodes the user prompts and prompts the mask decoder to generate segmentation masks. Unlike conventional SAM^{34,35}, which relies on visual prompts (e.g., points or bounding boxes), the SAM in UniBiomed is based on the textual prompts generated by MLLM. Following previous methods^{38,39,51}, UniBiomed combines the output of the MLLM with the tokenized user instructions as the language embeddings and injects them into SAM's prompt encoder, enabling text-guided segmentation. Specifically, the user instructions are first encoded by the pre-trained text tokenizer (i.e., BERT⁷⁸) and forwarded by the LLM. The LLM uses the standard new-token-prediction strategy to generate new tokens following the instructions. Then, we concatenate the embeddings of these generated tokens with the embeddings of the instructions, and inject them into the prompt encoder of SAM for prompting the SAM model. This effectively bridges the MLLM and SAM for grounded interpretation^{38,39}. Notably, following the previous methods^{38,39}, we adopt a special token “[SEG]” to instruct the mask decoder to produce segmentation masks. While for the tasks that do not require segmentation predictions, e.g., ROI classification and region-aware report generation, this special token is discarded, and the model will not produce segmentation predictions.

Notably, the success of UniBiomed stems from the complementary integration of MLLM and SAM. First, unlike the original SAM^{34,35}, which relies on visual prompts, UniBiomed employs language embeddings as textual prompts to guide SAM's mask decoder. This innovation eliminates the need for manually crafting precise bounding boxes—a major bottleneck in segmenting biomedical images with dense, irregularly shaped objects²². Second, segmentation training in UniBiomed implicitly enhances the diagnostic accuracy of the MLLM^{38,39,51}. Since segmentation predictions depend on the MLLM's textual prompts, the back-propagation process propagates supervision signals from the segmentation masks back to the MLLM, refining its ability to generate precise prompts. This synergistic integration not only improves segmentation performance but also strengthens the MLLM's diagnostic capabilities, creating a mutually reinforcing loop between the two components.

Following the state-of-the-art architecture Sa2VA³⁸, we adopt InternVL2.5³³ as the foundation MLLM and SAM2-hiera-large³⁵ as the segmentation model. To preserve the learned knowledge of the strong pre-trained MLLM, we leverage LoRA⁸⁰ to perform efficient fine-tuning of LLM, and freeze the vision encoder of MLLM entirely. For SAM2³⁵, following previous settings^{38,39}, we freeze the vision encoder and fine-tune the prompt encoder and mask decoder only.

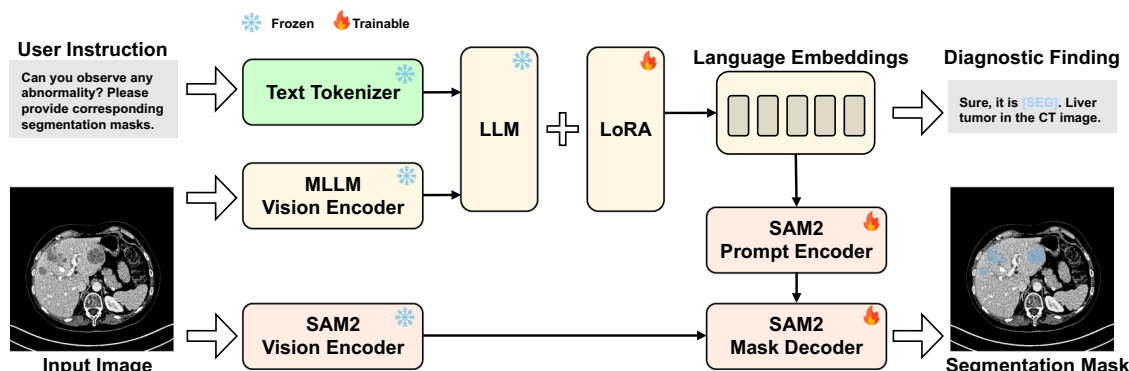


Fig. 6 | The architecture of UniBiomed. Specifically, the user instruction is processed by a pre-trained text tokenizer BERT⁷⁸ and input to LLM. The input image is encoded by the vision encoder of MLLM, and then the vision tokens are also injected into LLM for generating the diagnostic findings. Following previous methods^{38,39,51}, we combine the output of the MLLM with the tokenized user

instructions as the language embeddings and inject them into SAM2's prompt encoder for producing segmentation masks. Notably, we use LoRA⁸⁰ to finetune the MLLM as previous methods^{38,39}. In SAM2, the prompt encoder and mask decoder are fine-tuned, while the vision encoder is frozen. The used image is from¹¹.

We aggregate different sources of datasets into a universal training process. The overall training loss consists of a text generation loss L_{text} and a segmentation loss L_{seg} . The text generation loss L_{text} is an auto-regressive cross-entropy loss as standard LLM^{32,38,39}. The segmentation loss L_{seg} is a combination of per-pixel binary cross-entropy (BCE) loss L_{BCE} and typical dice loss L_{dice} . We balance the weights of loss functions, and the overall loss L is defined as:

$$L = L_{\text{text}} + L_{\text{seg}}, \quad L_{\text{seg}} = \lambda_{\text{BCE}} * L_{\text{BCE}} + \lambda_{\text{dice}} * L_{\text{dice}}. \quad (1)$$

Following previous methods^{22,38,39}, the coefficient of loss functions λ_{BCE} and λ_{dice} are defined as 2 and 0.5, respectively. The per-pixel binary cross-entropy (BCE) loss L_{BCE} is defined as:

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (2)$$

where $y_i \in \{0, 1\}$ is the ground truth label, $p_i \in [0, 1]$ is the predicted probability, and N is the total number of pixels. The dice loss L_{dice} is given by:

$$L_{\text{dice}} = 1 - \frac{2 \sum_{i=1}^N p_i y_i + \epsilon}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i + \epsilon} \quad (3)$$

where ϵ is a small constant for numerical stability.

Specifically, for the ROI classification and region-aware report generation tasks, the L_{seg} will be discarded during training. In this case, the loss function L is formulated as:

$$L = L_{\text{text}}. \quad (4)$$

Implementation

Pre-processing 3D medical images (CT/MRI). For pre-processing, we simply follow the settings of BiomedParse²² for fair comparisons, which are also widely adopted in recent medical vision-language models^{36,81,82}. Specifically, we first conduct normalization on the 3D images, i.e., window-adjustment for CT and Z-score normalization for MRI. In window-adjustment, we set the clipped Hopfield Units (HU) as $[-1000, 500]$ for chest CT and $[-175, 250]$ for abdomen CT. After normalization, we crop the 3D images along the axial axis to get the 2D slices.

For dataset curation, we adopt some readily available toolkits from SA2VA³⁸, RadGenome^{53,54}, VoCo^{83,84}, BiomedParse²², and MedTrinity³⁶ to transform the datasets into a uniform VQA format. Some examples are shown in Supplementary Fig. 4. For fair comparisons, we resize all images to 1024×1024 sizes for training and validation. We use Pytorch⁸⁵ to conduct all training and validation experiments. The training of UniBiomed is conducted on $8 \times \text{NVIDIA H800 (80G)}$ GPUs for 10 epochs, which takes about 5 days to finish the training. All the inference tasks can be established within one NVIDIA 3090 GPU (24GB). The seed is set to 42. We provide the available code links of comparison methods in Supplementary Table 26. More details of training and validation are shown in Supplementary Table 27.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

This study incorporates a total of 84 datasets across 10 diverse biomedical imaging modalities. All these datasets are publicly available for research. The details about the data used in this project are listed in Supplementary Tables 2, 3, 4, and 5. Source data are provided with this paper.

Code availability

The codes, datasets, and models of UniBiomed are available at GitHub (<https://github.com/Luffy03/UniBiomed>).

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Author contributions

L.W. and H.C. conceived this work. L.W. designed the framework and conducted the experiments. Y.N., S.H., J.Z. and L.L. contributed to the data acquisition and provided suggestions on the framework. T.L., Z.X., and D.C. contributed to the reader studies. Y.Z., N.M., V.V. and R.C.K.C. contributed to the evaluation and analyzed the results of UniBiomed. Y.P. and P.R. polished the manuscript and provided suggestions on the experiments. H.C. supervised the project. All authors discussed the results and contributed to the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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